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324635(24)

BE (6th Semester)

Examination, April - May, 2016

(Old Scheme)

Testing and Commissioning of Electric Equipments

Time Allowed : 3 hours *Maximum Marks* : 80

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Note : (*i*) Part (*a*) of each question is compulsory and solve any two parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Explain the difference between power transformer and distribution transformer. 2

(*b*) Enumerate various tests before commissioning of a power transformer. 7

(*c*) Explain the method for checking of the following : 7

(*i*) Polarity test of transformer winding

(*ii*) Ratio test

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(*d*) What are conditions of parallel operation of two power transformers? Explain how percentage impedance effects the operation of two parallel connected transformers. 7

2. (*a*) Write two applications of rectifier transformer. 2

(*b*) What are the major differences in design of traction transformer compared to power transformer? 7

(*c*) Explain the principle of operation of transductor. 7

(*d*) Explain how magnetisation curve of CT is obtained. Explain the effects of saturation of performance of CT. 7

3. (*a*) What is the function of brushes and commutators in DC machines? 2

(*b*) Which are the tests to be conducted on large induction motors before commissioning? 7

(*c*) What are the reasons of noise in induction motors and how noise can be reduced? 7

(*d*) Explain the factors affecting the life of commutator, brushes and slip ring in DC machine. 7

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4. (*a*) Define polarization index. 2

(*b*) What are the main steps in drying out process? Describe self resistance method to measure temperature of winding of rotating machine. 7

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(*c*) Describe the various classes of standard operating duty of a rotating machine. 7

(*d*) Specify the different types of enclosures of motors, highlighting the special feature of each type. Mention the area of application of each type. 7

5. (*a*) Define clearance in an electrical system. 2

(*b*) What is Equipment Earthing? How does this provide safety to operator? Explain. 7

(*c*) What are the first aid measures taken at site when a person gets electric shocks? 7

(*d*) State the seven principles of safety management. 7

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